MC 501 Describing Data in R rubric

100 points. Due Monday, November 23, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- The calls these tables and figures lean on, all of which run the same in VS Code: `count()` for a frequency table, `pivot_wider()` to lay a cross-tabulation out, `fct_reorder()` to order the bars, `labs()` for the axis labels, title, and caption, `scale_fill_brewer()` or the v2v palette for the fill, and position = "fill" with `scales::percent_format()` for a proportion chart.
- Recommended, not graded here: open the script with the settings block Inferencing Data requires (load the data, rename the variables you use to a fixed set, and make the categorical ones factors), so that next week's script is this one with four tests added.
- What it feeds: your White Paper, which runs these same table and figure calls on your own variables. The Twitch figures stay here; the paper shows only what your own data shows.

Describing Data in R, 100 points

Criterion	Exemplary	Proficient	Developing	Beginning
Tables and the three numbers 20 points	**20 points.** Frequency tables for the key categorical variables, built with `count()`, and the mean, median, and standard deviation for every continuous variable you report.	**15 to 19 points.** All there, but a table is unlabeled or one variable is missing a number.	**10 to 14 points.** The mean alone, or tables for only some variables.	**0 to 9 points.** None.
Figures 20 points	**20 points.** At least two ggplot2 figures plus one figure of your own choice, labeled, titled, and captioned, saved to a file by the code that makes them, with one sentence on why you chose the third figure for that variable.	**15 to 19 points.** The figures are there, but one is missing a label, the title, or the caption, or the figure you chose has no reason given.	**10 to 14 points.** One figure, or figures saved by hand instead of by the code.	**0 to 9 points.** No figures.
Cross-tabulation 10 points	**10 points.** A cross-tabulation of two categorical variables, laid out with `count()` and `pivot_wider()`, labeled so a reader can tell the rows from the columns. This is the table next week's chi-square tests.	**8 to 9 points.** Present, but the labels are missing.	**5 to 7 points.** Built on the wrong columns.	**0 to 4 points.** Missing.
Alt text 5 points	**5 points.** Every figure has alt text that says what the figure shows.	**4 points.** Alt text on every figure, but one repeats the caption.	**3 points.** Alt text on some figures only.	**0 to 2 points.** None.

Criterion	Exemplary	Proficient	Developing	Beginning
Diagnostics behind each figure 25 points	**25 points.** For each figure: the shape of the variable, whether the assumptions of the tests you are heading toward look reasonable, and where the outliers are, with the check shown.	**19 to 24 points.** Shape and outliers reported, but the assumptions are not tied to the tests you plan to run.	**13 to 18 points.** Diagnostics for some figures only, or stated without being shown.	**0 to 12 points.** Figures without diagnostics. The figure is the claim; the diagnostic is whether you are allowed to make it.
Interpretation 15 points	**15 points.** Two to three sentences after each output.	**11 to 14 points.** Sentences after each output, but they describe the output without saying what it cannot show.	**8 to 10 points.** Interpretation for some outputs only, or a single line each.	**0 to 7 points.** None.
Display choices disclosed 5 points	**5 points.** Every display choice disclosed: bin width, axis caps, merged categories.	**4 points.** One choice is not disclosed.	**3 points.** One disclosure across all the figures.	**0 to 2 points.** None.